

A cofactor row by row replacement

MathlibAnnex.Piola.cofactorRow

def

Fixes the cofactor convention used in the divergence and determinant identities.

Statement

Let $A = (A_{kq})$ be a real $n \times n$ matrix, and fix a row index $i \in \{1, \dots, n\}$. Write $e_j = (\delta_{jq})_{q=1}^n$ for the j th coordinate vector. The cofactor row $C_i(A)$ is the vector whose j th coordinate is the determinant obtained by replacing row i of A by e_j^T .

Definition

For a row vector v , let $A[i \leftarrow v]$ denote A with only its i th row replaced by v . Then

$$C_i(A) = (C_i(A)_j)_{j=1}^n, \quad C_i(A)_j = \det(A[i \leftarrow e_j^T]).$$

Equivalently, $C_i(A)_j$ is the usual (i, j) cofactor. The row-replacement formula fixes the row and column convention without choosing an inverse.

Assumptions

The matrix is arbitrary: invertibility is not required. The dimension is a natural number, with a specified $i \in \{1, \dots, n\}$. When $n = 0$ there is no such index, so this indexed definition has no instance to evaluate.

Conclusion

The result is a row in $\mathbb{R}^n$, using the row convention throughout. No transpose or choice of an inverse is part of the definition.

The later row-expansion lemma gives $\det(A[i \leftarrow v]) = \sum_j v_j C_i(A)_j$. It is cited as a separate identity, not included as an additional conclusion of this definition.

Main citations

- Exact declaration and its source — MathlibAnnex.Piola.cofactorRow
- Expansion along a replaced row — MathlibAnnex.Piola.det_updateRow_eq_sum_mul_cofactorRow
- Coordinate row — MathlibAnnex.Piola.basisRow
- Entries of a coordinate row — MathlibAnnex.Piola.basisRow_apply

Lean source signature (exact)

```
def cofactorRow {n : ℕ} (A : Matrix (Fin n) (Fin n) ℝ)
  (i : Fin n) : (Fin n → ℝ) :=
  fun j => (A.updateRow i (basisRow j)).det
```

In the source	Mathematical meaning
$\{n : \mathbb{N}\} (A : \text{Matrix } (\text{Fin } n) (\text{Fin } n) \mathbb{R}) (i : \text{Fin } n)$	An arbitrary real $n \times n$ matrix A and the fixed row i . No invertibility hypothesis occurs; for $n = 0$ no row index can be supplied.
$: (\text{Fin } n \rightarrow \mathbb{R})$	The output cofactor row $C_i(A) \in \mathbb{R}^n$.
<code>basisRow j</code>	The coordinate row $e_j^T = (\delta_{jq})_q$, in the cited definition.

In the source	Mathematical meaning
<code>A.updateRow i (basisRow j)</code>	The matrix $A[i \leftarrow e_j^T]$: i is the replaced row and j is the output cofactor coordinate.
<code>fun j => (A.updateRow i (basisRow j)).det</code>	The entire RHS $C_i(A)_j = \det(A[i \leftarrow e_j^T])$ at each j . It does not introduce an inverse or transpose of the resulting row.

▼ Exact surrounding binder context (separate excerpt)

```
noncomputable section

open scoped BigOperators

namespace MathlibAnnex
namespace Piola
```

▼ Exact content identity

Declaration: `MathlibAnnex.Piola.cofactorRow`

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